Strategic Valuation of Public Housing: A Machine Learning Approach to Singapore HDB Resale Prices

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Group 7 Presentation

Cultural, MRT Station & Primary School Proximity Influence

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AGENDA

Strategic Valuation of Public Housing — A Machine Learning Approach

How can we build a reliable and explainable model to estimate Singapore HDB resale value using the factors that truly matter, while separating real pricing signals from noise and market myths?

Why Enrich the Data?

The Problem

The raw HDB Resale dataset from data.gov.sg has only 11 columns — mostly flat attributes (type, area, storey).

Location is captured only as "town" and "street_name" — coarse text labels with no geospatial precision.

Without these features, the model cannot learn location-dependent pricing patterns.

The existing dataset does not have two critical factors that drive property prices: proximity to transport hubs (MRT/LRT) and primary schools.

The Solution

Enrich each transaction with geo-location using OneMap for:

- Top 3 nearest MRT/LRT stations
- Nearest primary school
- Lunar Month

Transforms coarse location text into continuous numerical features for ML models.

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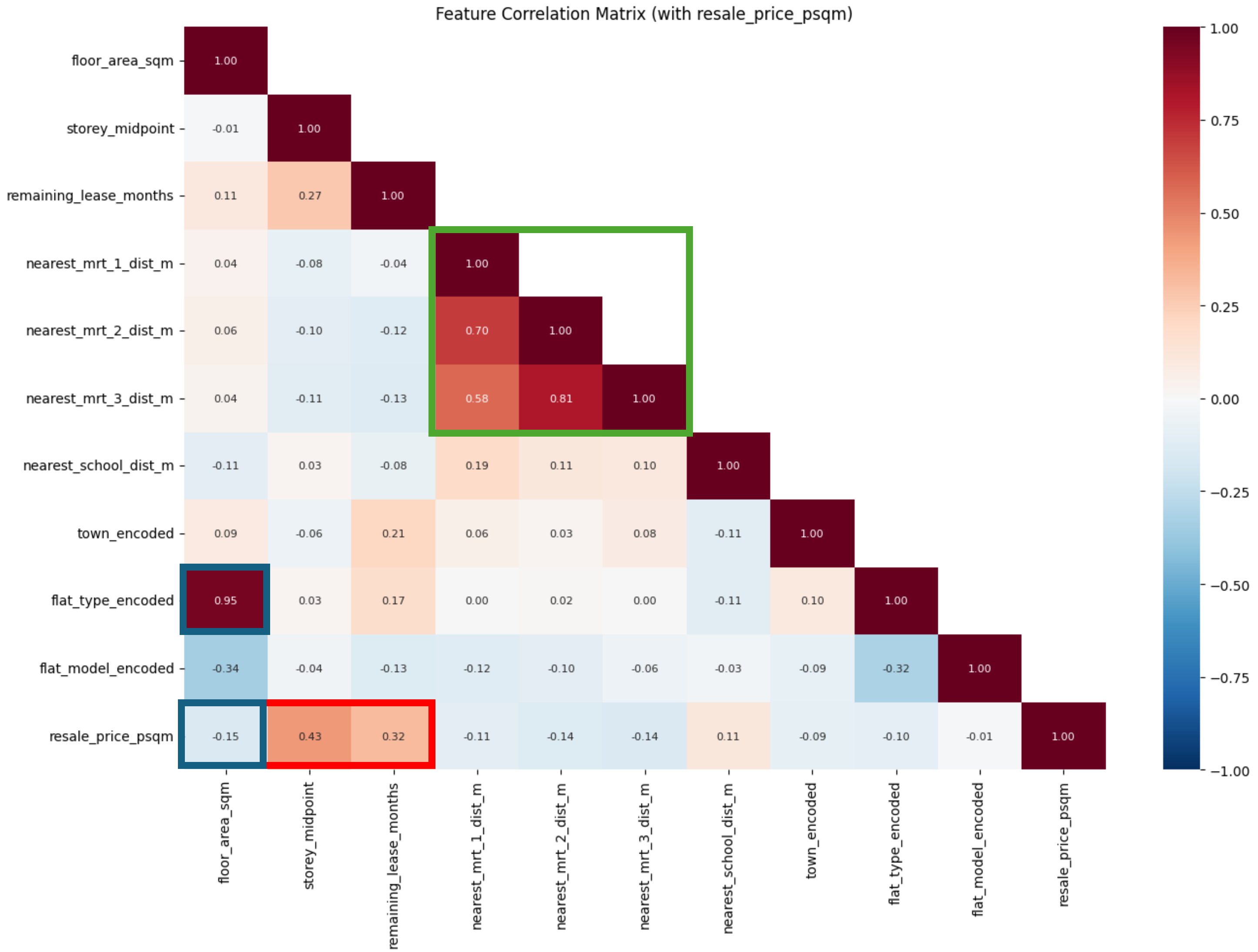
Exploratory Data Analysis: Correlation Heatmap

storey height and remaining lease are the strongest linear correlates



Correlation Heatmap:

- storey_midpoint (r=+0.43) and remaining_lease_months (r=+0.32) are the strongest positive correlates with resale_price_sqm.
- floor_area_sqm (r=−0.15) is NEGATIVE - the economy-of-scale signature.
- flat_type_encoded is the ordinal encoding of flat type, where 1 ROOM = 1, 2 ROOM = 2, … up to MULTI-GENERATION = 7. It captures the size category of the flat as defined by Housing Development Board (HDB).

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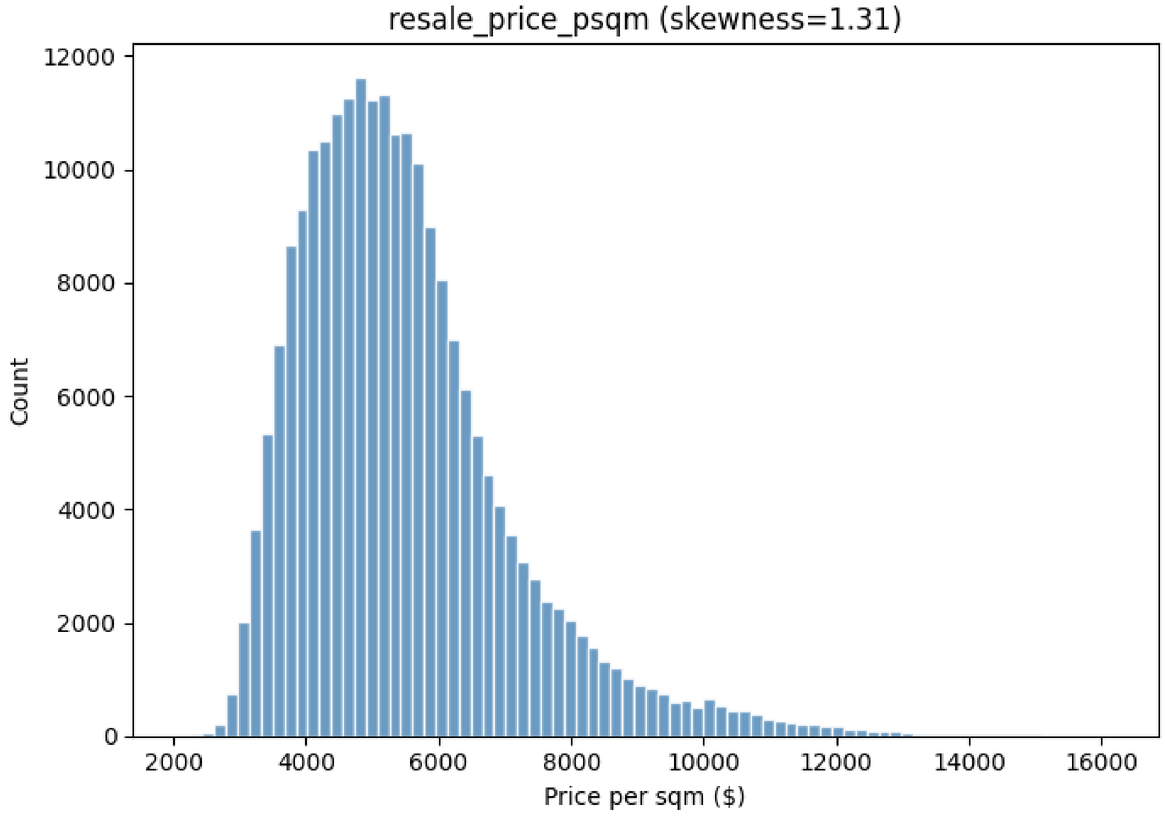
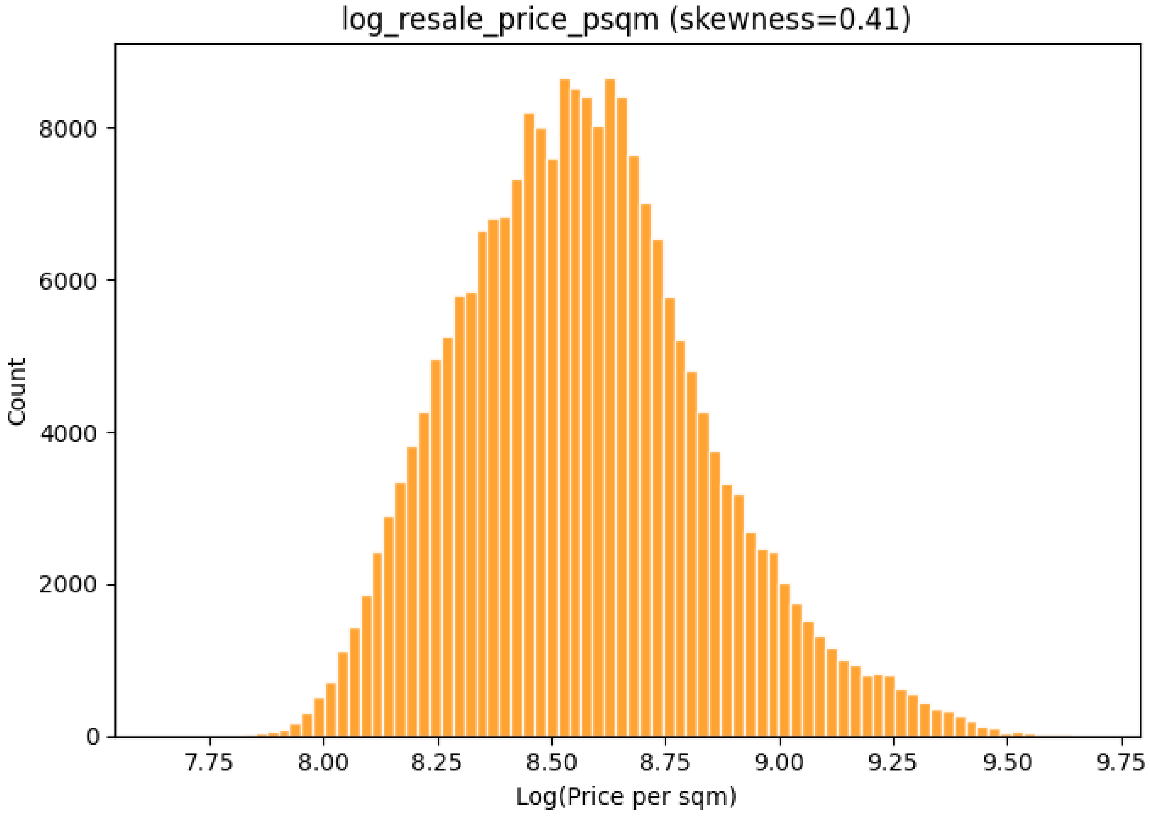
Exploratory Data Analysis:

Price Distribution: Raw vs Log-Transformed psqm

Log-transform is justified by skewness reduction from 1.31 → 0.41





⚠ Right-skewed: Tail inflated by premium units in Queenstown, Bukit Timah, Central Area. Raw skewness = 1.31.

✓ Near-Normal after log-transform: Skewness falls to 0.41. Symmetric distribution suits regression and Gaussian Naïve Bayes (GNB).

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Geocoding 224K transactions, MRT/LRT, Schools distance features & 8-phase pipeline

2 **Event Regime Study**
Impact of covid, cooling measures and rate hike

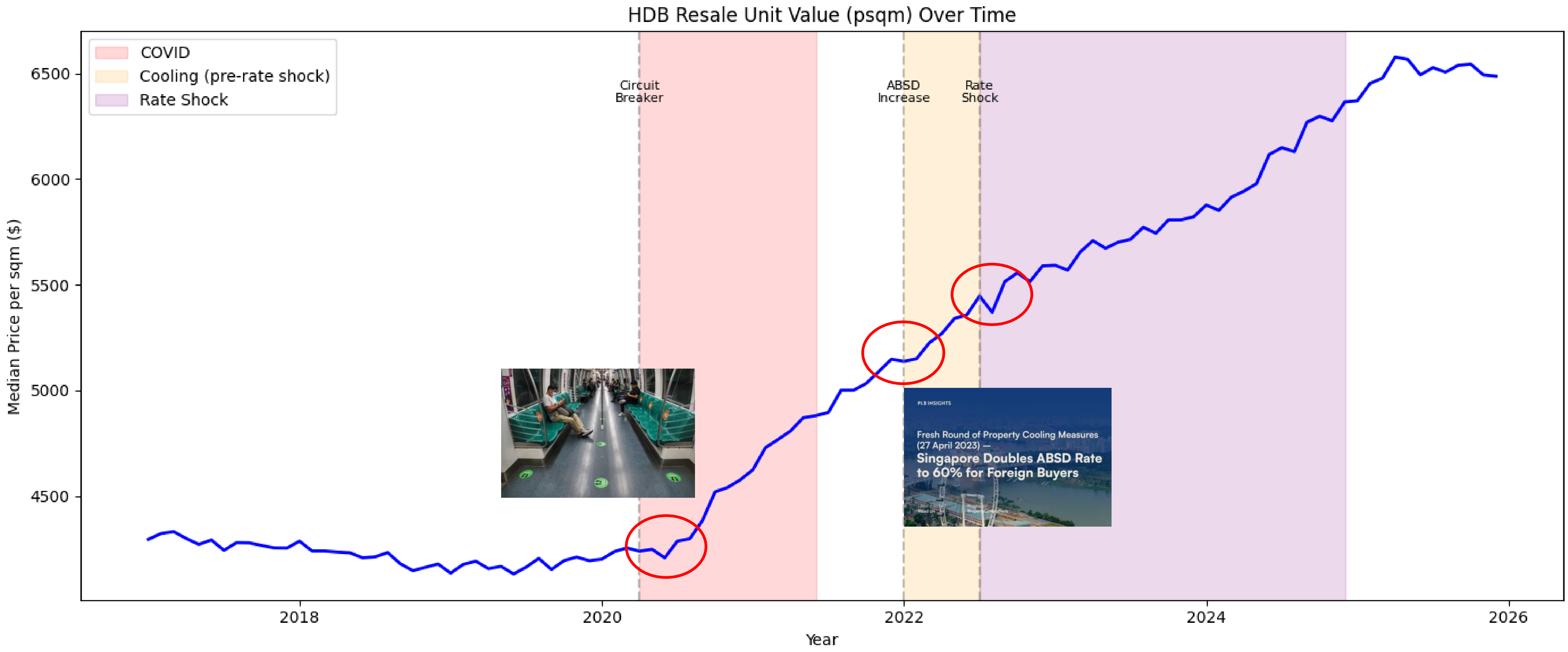
3 **School Proximity Premium**
Good-deal classification via stratified median & Mutual Information analysis

4 **MRT, Storey & Lease — Price Drivers**
Correlation analysis, EDA fly-bys & linear regression feature selection

5 **Best Price Estimator — Ensembles**
Random Forest, Gradient Boosting, & Stacking

6 **Conclusion & Key Findings**
Regime impact summary, model recommendations & practical implications

How do structural events, i.e. COVID-19, property cooling, and interest rate shocks, affect HDB resale price?



Pre-COVID (before Apr 2020)	COVID (Apr 2020 – Jun 2021)	Cooling (Jan 2022 onwards)	Rate shock (Jul 2022 -- Dec 2024)

What affects did Covid have on HDB prices?
Analyzing the "New Normal" in HDB Resale Trends

- **Pre-COVID vs. Post-COVID Shifts:** Significant price acceleration observed post-2020 due to construction delays in the BTO (Build-To-Order) market.
- Analysis of how "Work From Home" trends shifted buyer preference toward larger flat types (4-room and 5-room) and higher floor levels.
- **Feature Sensitivity: Lease Decay:** During the pandemic, the "remaining lease" feature became a stronger price driver as buyers sought long-term stability.
- **Proximity to Amenities:** While MRT proximity remained a "Gold Standard" feature, the relative weight of "Primary School Proximity" saw a premium spike in specific high-demand towns.
- **Policy & Economic Cooling:** Visualizing the interaction between COVID-era low interest rates and subsequent cooling measures introduced to stabilize the steep price climb.

Three COVID-Era Sub-Periods (each with own train / valid / test & own price tier thresholds)

Pre-COVID (2017–19): Premium threshold $4,624 psqm **During-COVID (2020–21):** Premium threshold $4,785 psqm
Post-COVID (2022–25): Premium threshold $6,053 psqm

13 Features Used

MRT-1/2/3 distance, School distance, School <1km, Flat type, Town, Floor level, Floor area, Remaining lease months, Year, Lunar month, MRT access tier

Log-transform applied to psqm (skewness 1.31 → 0.41) before tiering

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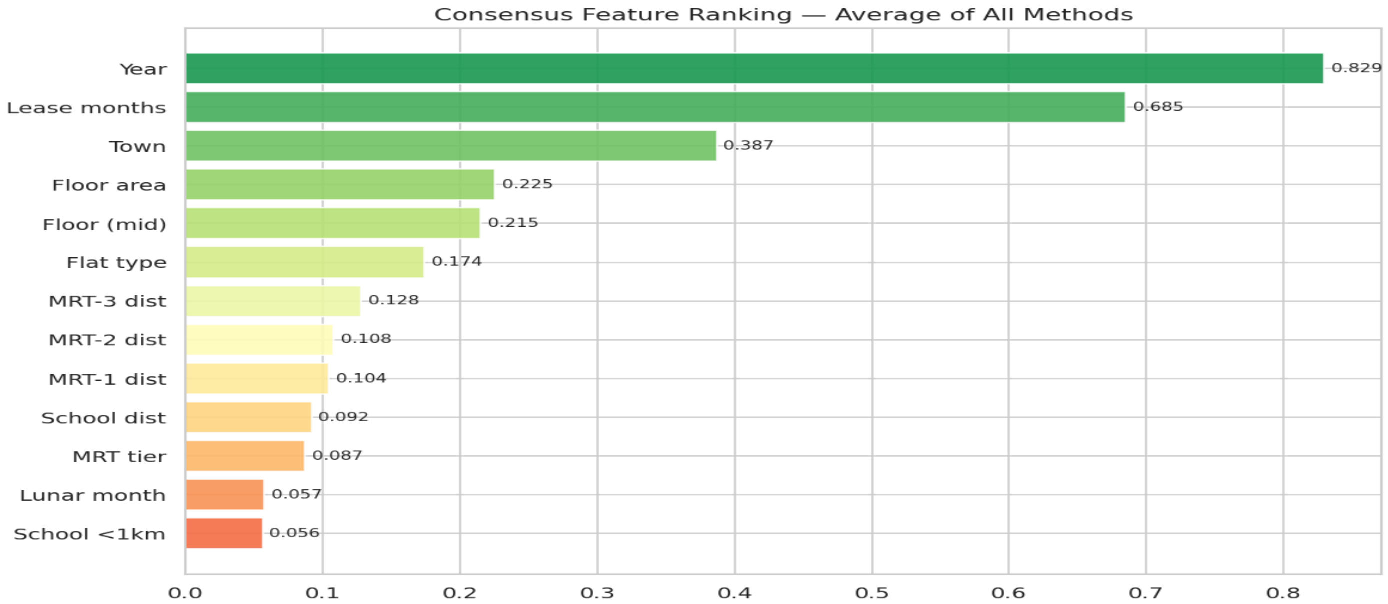
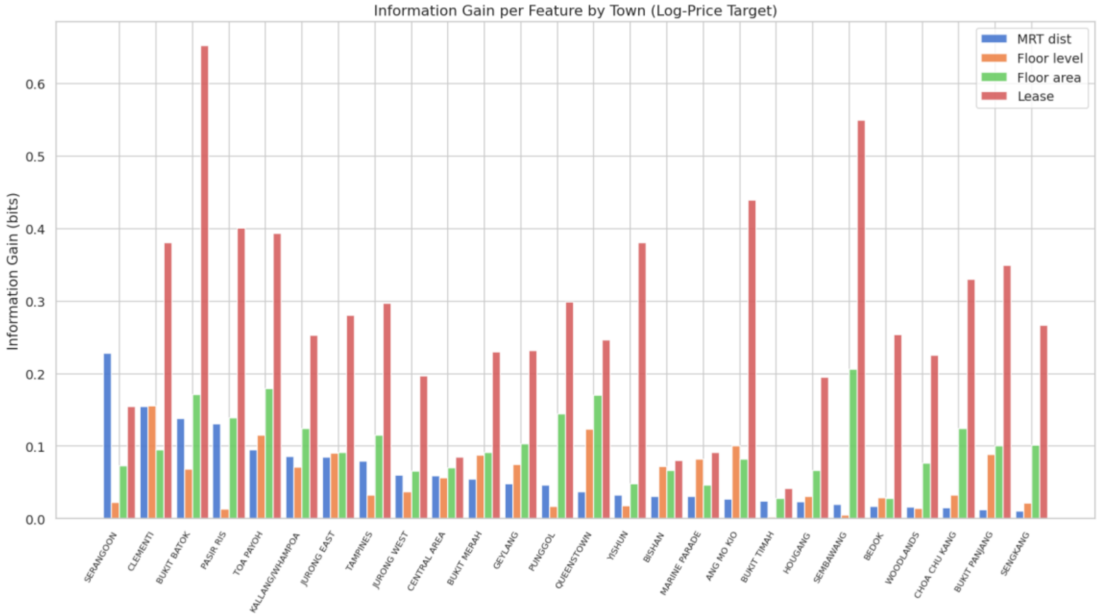
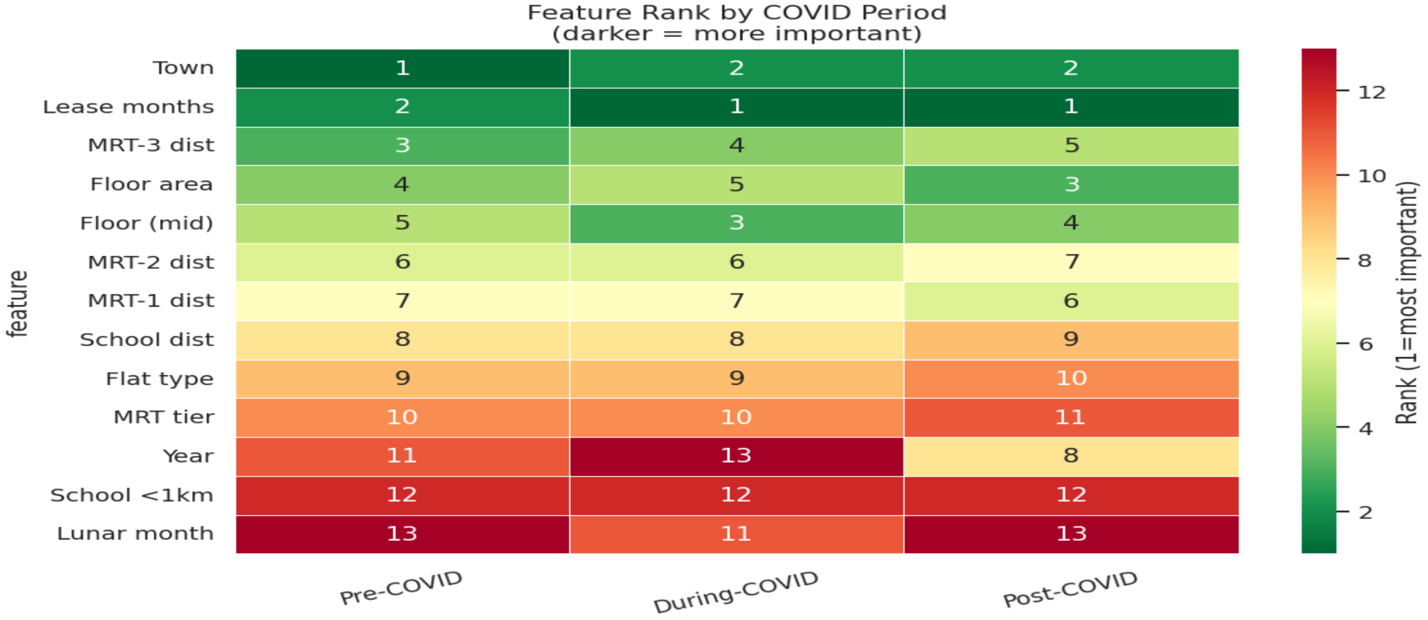
Factor Ranking

- **Key Price Drivers Identified:**
- **Storey Range:** High-floor units consistently commanded a premium, especially in mature estates.
- **Remaining Lease:** A critical feature for calculating the real value of older HDB units.
- **MRT Proximity:** Quantifying the "walking distance" premium (e.g., 500m vs 1km).
- **Mutual Information Analysis:** Using statistical methods to rank which features (like town_premium vs flat_model) shared the most information with the target price.







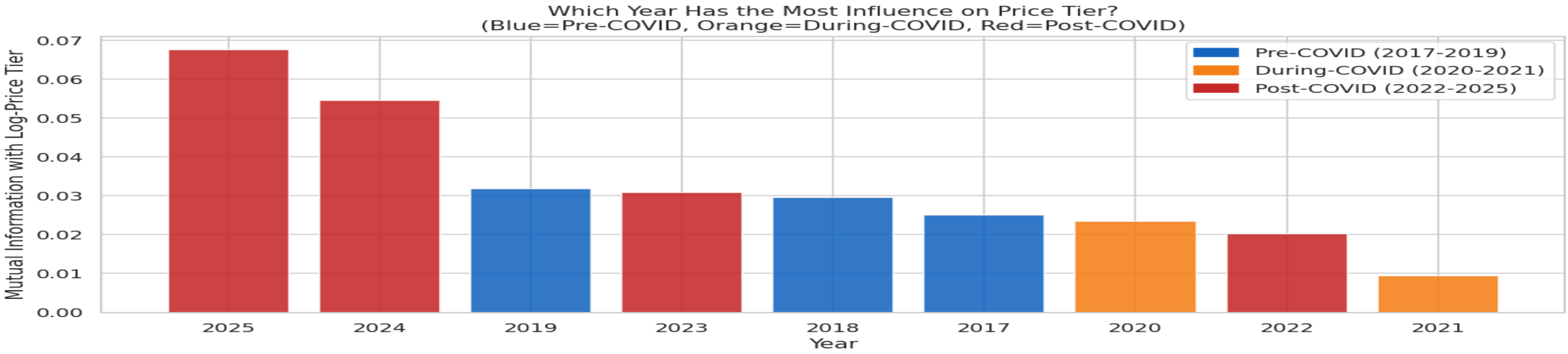
RQ1: Key Findings — Structural vs Transient Pricing Drivers

01
Lease & Town are structural
Remaining lease months and Town rank #1–2 across all 9 methods and hold their rank in the 2025 test set. Property agents can rely on these rules across market cycles.

02
Year is transient, to be avoided in deployment
Year ranks #1 by DT importance (0.446) on training data but collapses to zero IG in 2025. It acts as a market-regime proxy — useful in-sample, unreliable out-of-sample.

03
MRT premium is regime-sensitive
MRT ranks 3–4 in training but becomes less discriminating post-COVID. Walkable MRT (<300 m) retains premium; the 600 m–1 km band saw the steepest relative decline during WFH.

04
Lunar month: near-zero signal
Lunar month ranks last (#13) across all 9 methods. IG never exceeds 0.018 bits in any year. It provides no meaningful incremental information once structural features are not accounted.



Consensus rank (avg of 9 methods): Year 2.44 ● Lease 2.89 ● Town 4.22 ● Floor level 5.22 ● Floor area 5.44 ● MRT distances 6–9 ● Lunar month 10.9

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Research Question 2

RQ2: Predicting whether HDB flats command a school premium

Research Design

Target Variable

school_premium = 1 if flat is within 1km of a ranked school (school_rank not null), else 0

→ 32,162 positive / 192,503 negative (14.3% positive)

Key Design Choice

Stratified 5-Fold CV used instead of temporal split, school proximity is a geographic/structural relationship, not time-dependent

Leakage Prevention

Excluded: nearest_school_dist_m (used to build target), nearest_school (name), school_rank (directly defines label)

Features Used

Numeric: resale_price_psqm, floor_area_sqm, storey_midpoint, remaining_lease_months, MRT distances (×3)

Categorical: town, flat_type, flat_model

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Classifier CV Results (Stratified 5-Fold)

Model	F1	ROC-AUC	Precision	Recall
Logistic Regression (L2)	0.5711 ± 0.0037	**0.9085 ± 0.0024**	0.4353 ± 0.0026	0.8300 ± 0.0077
Logistic Regression (L1)	0.5709 ± 0.0038	0.9083 ± 0.0025	0.4351 ± 0.0027	0.8298 ± 0.0078
Decision Tree (depth=5)	0.8698 ± 0.0195	0.8563 ± 0.0064	0.9800 ± 0.0015	0.7825 ± 0.0323

0.91

ROC-AUC

Strong class separation

0.83

Recall (LR)

Catches most premium flats

14.3%

Positive class

Real imbalance handled by class_weight='balanced'

High ROC-AUC (0.91) shows strong separation. High recall but low precision reflects class_weight='balanced' trading precision for minority class coverage.

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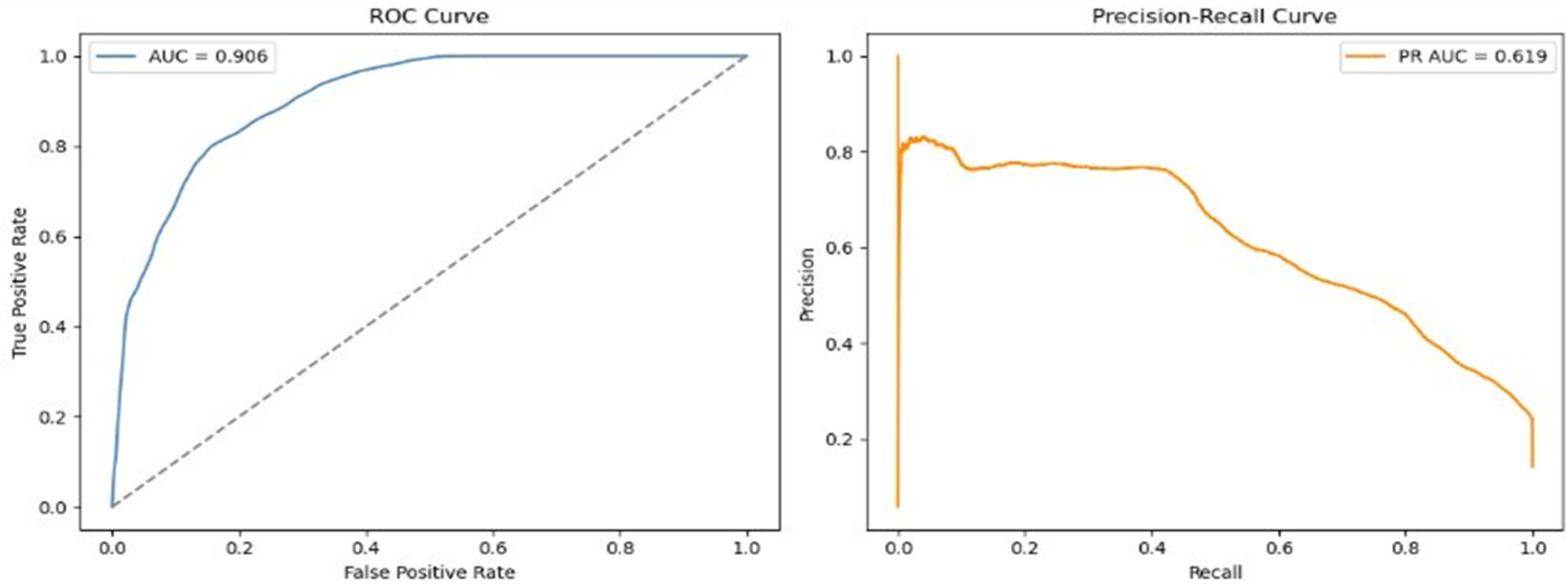
ROC & Precision-Recall Curves



AUC = 0.906:

Model correctly ranks premium flats above non-premium in 91% of cases

PR AUC = 0.619:

Moderate precision degrades past recall ~0.5, revealing the imbalance challenge

Logistic Regression Coefficient Analysis

Most predictive of school premium

- Geylang (5.82)
- Marine Parade (5.42)
- Toa Payoh (5.24)
- Bukit Timah (4.67)
- Kallang/Whampoa (3.71)
- Central Area (3.00)
- Bishan (2.74)
- DBSS flat model (2.63)

Least predictive of school premium

- Pasir Ris (-8.30)
- Jurong West (-6.80)
- Woodlands (-6.50)
- Ang Mo Kio (-5.81)
- Jurong East (-5.60)
- Sembawang (-5.35)
- Clementi (-5.15)
- Bukit Panjang (-4.99)

Key takeaway: School premium is overwhelmingly determined by town/location. Mature, central estates (Geylang, Marine Parade, Toa Payoh) cluster near ranked schools due to Singapore's urban planning history. Peripheral new towns (Pasir Ris, Jurong West) have fewer ranked schools nearby.

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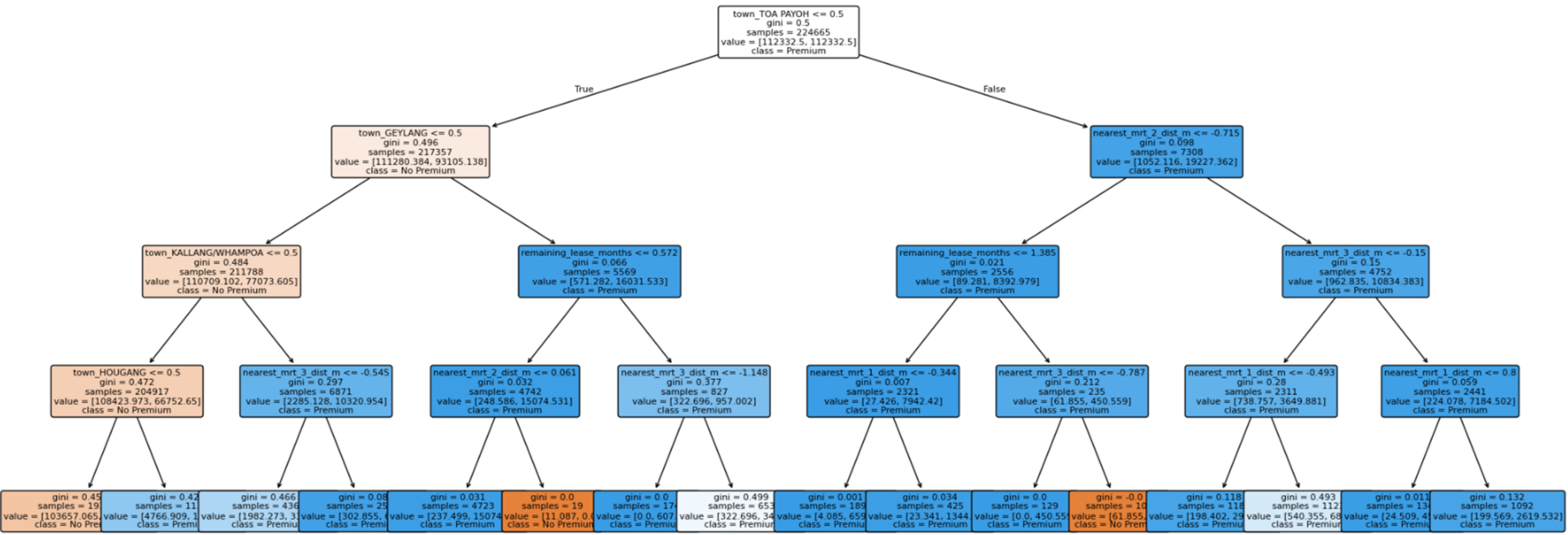
Decision Tree Visualisation

Decision Tree (max_depth=4)



Root split on town_TOA PAYOH

Location is the single most informative feature, confirming coefficients

Left branch: mature central towns

Geylang, Kallang/Whampoa, Hougang, geographic clustering of ranked schools

Right branch: MRT distance secondary

In non-central towns, transport access acts as proxy for urban density

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Polynomial Ridge & Lasso Regression (degree=2)

Target: log(resale_price_psqm) — log-transformed to address skewness

Model	R^2 (test)	RMSE (log scale)	Interpretation
Ridge (α=1.0)	**0.6645 ± 0.0029**	0.16 ± 0.00	~17% avg error in price terms
Lasso (α=1.0)	-0.0001 ± 0.0001	0.27 ± 0.00	Effectively failed , no better than mean

Ridge wins clearly

R^2 of 0.66 vs Lasso's -0.0001. Most features contribute, Lasso's zeroing is too aggressive with degree-2 polynomial features.

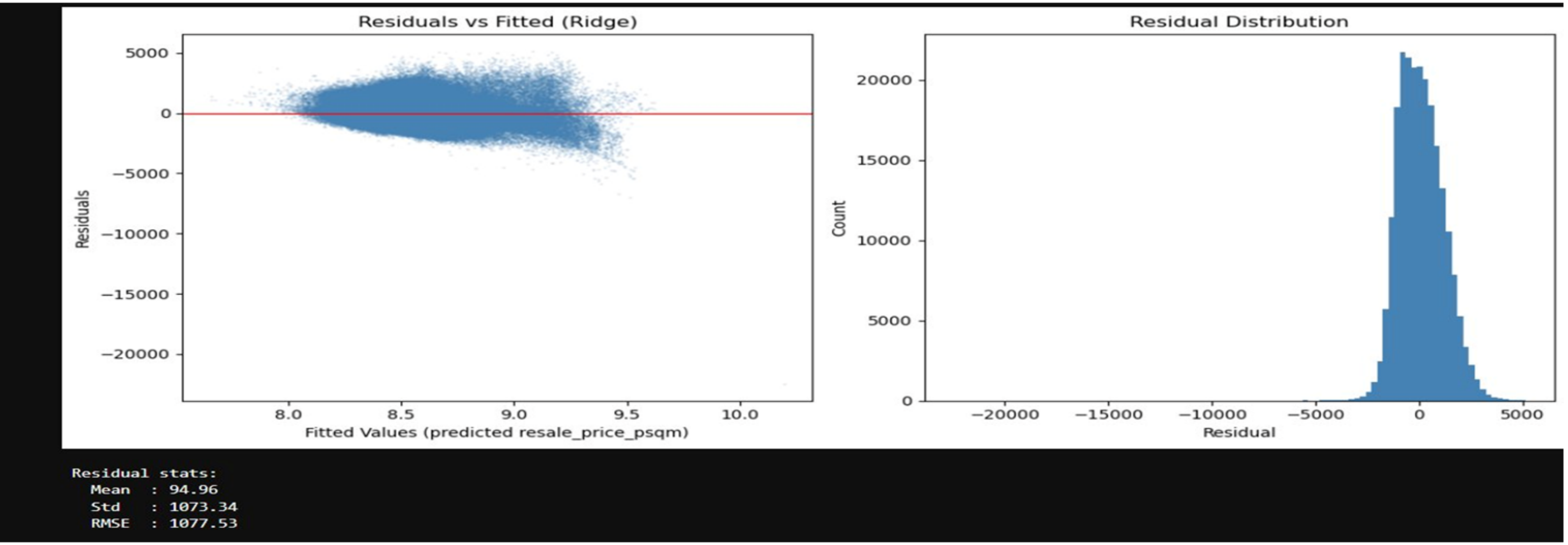
66% variance explained

Town, flat type, storey, lease, and MRT distance collectively explain 66% of log-price variance. Remaining 34% reflects renovation, timing, and buyer-specific factors.

Log transform applied

As advised, log(resale_price_psqm) was used as target to address skewness. RMSE of 0.16 log units ≈ 17% error in real price terms.

Residual Analysis (Ridge Regression)



Mean
94.96
Near-zero: largely unbiased

Std
1,073.34
Spread in $/sqm

RMSE
1,077.53
~20% of typical price range

Heteroscedasticity persists after log-transform — spread widens for high-value flats. Left-skewed distribution indicates occasional over-prediction. A non-linear model (e.g. gradient boosting) may better handle variance instability.

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Key Findings

01 Location dominates

Town is the single strongest predictor, mature central estates (Toa Payoh, Marine Parade, Geylang) are most proximate to ranked schools.

02 Strong discrimination

ROC-AUC of 0.91 confirms the classifier reliably separates premium from non-premium flats despite 14.3% class imbalance.

03 Ridge >> Lasso

Ridge (R^2=0.66) vastly outperforms Lasso (R^2≈0) for price regression, most polynomial features contribute meaningfully, Lasso over-penalises.

04 Residual limitations

Heteroscedasticity in residuals suggests unobserved factors (renovation, buyer premiums) drive error at the high-price end, a gradient boosting model may help.

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RQ3: Understanding HDB Price Drivers

Linear Regression · Feature Engineering · Regularisation · Feature Selection

What are the structural relationships between MRT proximity, storey height, remaining lease, and HDB resale price per sqm?

Q1
Is the MRT effect non-linear?

Q2
Does polynomial expansion + regularisation improve fit?

Q3
Which features does Lasso select as most important?

Numeric Variables (7)

Feature	Description
nearest_mrt_1_dist_m	Distance to nearest MRT
nearest_mrt_2_dist_m	Distance to 2nd nearest MRT
nearest_school_dist_m	Distance to nearest school
storey_mid	Midpoint of storey range
remaining_lease_months	Remaining lease in months
floor_area_sqm	Floor area (sqm)
gregorian_year	Transaction year

Categorical Variables (2)

Feature	Description
flat_type	HDB flat type (3-room, 4-room, etc.)
town	HDB town / planning area

Target: `resale_price_psqm`
SGD per square metre (log-transformed)

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Feature Engineering

Feature Extraction

New Feature	Source	Operation
`storey_mid`	storey_range	"10 TO 12" → midpoint → 11.0
`remaining_lease_months`	remaining_lease	"61 years 04 months" → 736 months
`nearest_mrt_2_dist_m`	original (missing)	Imputed as nearest_mrt_1_dist_m × 1.3

Scaling & Encoding

Transform	Applied To	Purpose
Log Transform	Target variable	Compress right-skewed distribution; satisfy LR assumptions
StandardScaler	All numeric features	Remove scale differences between features
OneHotEncoder	flat_type, town, regime	Encode categorical variables as binary columns

Polynomial Expansion

Transform	Applied To	Purpose
PolynomialFeatures(deg=2)	All features (post-encode)	Generate squared terms + interaction terms automatically
StandardScaler (2nd)	All poly-expanded features	Re-standardise to prevent squared terms from dominating

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Hyperparameter Search

Ridge α Search

α	Val RMSE	
0.001	485.2590	
0.01	485.2590	
0.1	485.2589	
1.0	485.2580	
10.0	485.2544	← Best
100.0	485.4286	
10000.0	500.6679	
1000000.0	1185.2702	

Best α = 10 (Val RMSE = 485.2544)

Lasso α Search

α	Val RMSE	
0.0001	**489.4458**	← Best
0.001	506.8341	
0.01	732.1540	
0.1	1486.0816	
1.0	1645.1126	
10.0	1645.1126	
100.0	1645.1126	
1000.0	1645.1126	

Best α = 0.0001 (Val RMSE = 489.4458)

Key insight: Ridge is insensitive to α (flat RMSE curve) — multicollinearity is not a major issue. Lasso's higher RMSE at large α shows aggressive feature elimination degrades fit.

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5-Fold Cross-Validation Evaluation

Model	CV R^2 (mean ± std)	CV RMSE (mean ± std)	CV MAE (mean ± std)
OLS (raw)	0.8545 ± 0.0017	618.7 ± 2.4	470.6 ± 1.6
OLS (poly deg-2)	**0.9106 ± 0.0009**	**485.1 ± 1.2**	**360.9 ± 0.8**
Ridge (α=10, poly)	**0.9106 ± 0.0009**	**485.1 ± 1.2**	**360.9 ± 0.8**
Lasso (α=0.0001, poly)	0.9083 ± 0.0034	491.1 ± 7.7	362.2 ± 0.5

0.9106
Best R^2
OLS poly / Ridge

485.1
Best RMSE
SGD/m²

360.9
Best MAE
SGD/m²

+6.6%
R^2 Improvement
vs raw OLS

Key findings: Polynomial features boost R^2 from 0.85 to 0.91. Ridge ≈ OLS (poly) confirms low multicollinearity. Lasso trades slight accuracy for 56% feature reduction.

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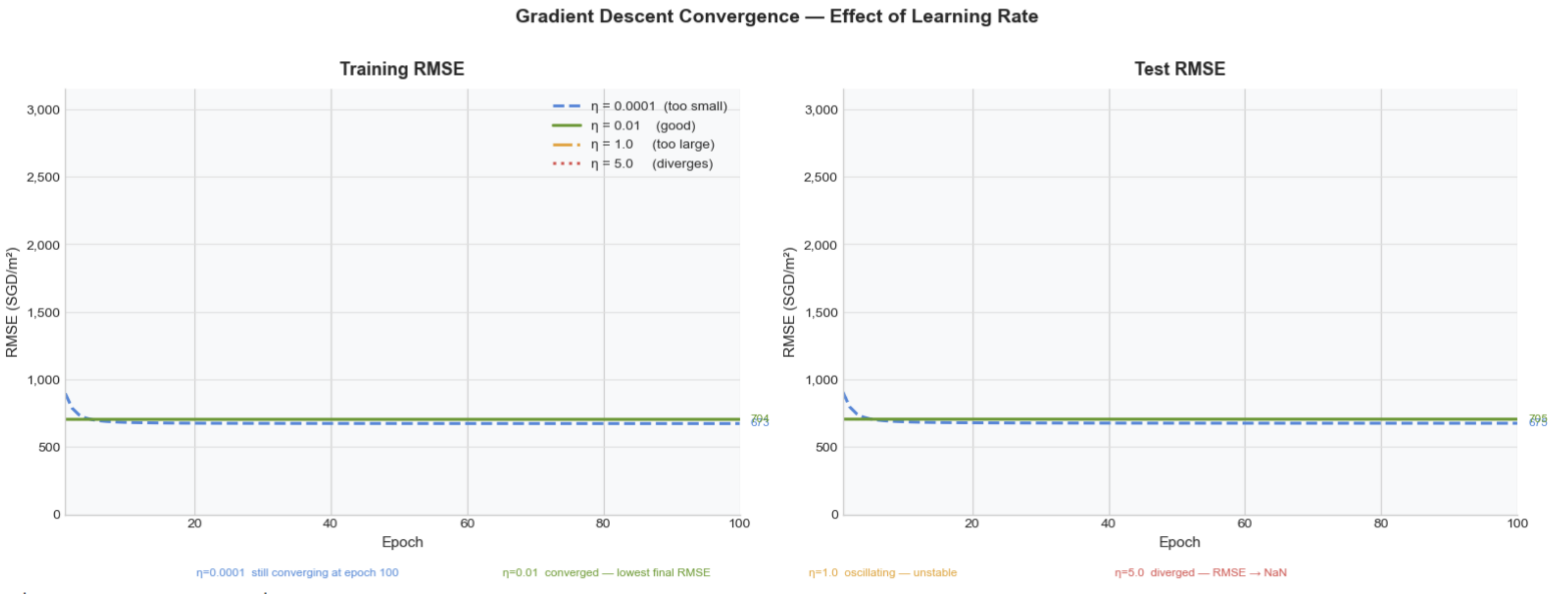
Gradient Descent Convergence



Learning Rate (η)	Train RMSE	Test RMSE	Status
0.0001	672.95	675.14	✓ Converged
0.01	**704.03**	**705.25**	✓ Converged
1.0	NaN	NaN	✗ Diverged
5.0	NaN	NaN	✗ Diverged

η = 0.0001 achieves lowest RMSE but is still converging at epoch 100. **η = 0.01 converges fast — best practical choice.**

Lasso Coefficient Analysis

Top 20 features by absolute coefficient value (α = 0.0001, polynomial deg-2)

860
Total Features · after poly expansion

485
Zeroed by Lasso · 56.4% eliminated

375
Surviving Features · 43.6% retained

#	Feature	Coeff
1	gregorian_year	+0.1540
2	remaining_lease_months	+0.0807
3	remaining_lease × floor_area	−0.0572
4	floor_area_sqm	−0.0510
5	town_BUKIT MERAH	+0.0490
6	storey_mid	+0.0481
7	remaining_lease × 5-ROOM	+0.0466
8	town_QUEENSTOWN	+0.0410
9	floor_area × town_PUNGGOL	+0.0382
10	town_CHOA CHU KANG	−0.0374

#	Feature	Coeff
11	town_BISHAN	+0.0368
12	town_WOODLANDS	−0.0354
13	nearest_mrt_1_dist_m	−0.0349
14	remaining_lease × 4-ROOM	+0.0322
15	gregorian_year²	+0.0315
16	5-ROOM × town_PUNGGOL	−0.0293
17	town_KALLANG/WHAMPOA	+0.0292
18	town_TOA PAYOH	+0.0290
19	floor_area × 3-ROOM	−0.0288
20	MRT2_dist × town_CENTRAL	−0.0288

Interpretation: gregorian_year (+0.154) is the dominant driver — HDB prices have risen over time. Town location and remaining lease are the next strongest

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Lasso Coefficient Analysis

Top 15 most stable features sorted by |Mean Coefficient| — 5-Fold CV stability check

#	Feature	Mean Coef	CV (%)
1	gregorian_year	0.154104	0.47%
2	remaining_lease_months	0.080577	0.38%
3	remaining_lease × floor_area	−0.057138	2.81%
4	floor_area_sqm	−0.051676	1.89%
5	town_BUKIT MERAH	0.048891	1.25%
6	storey_mid	0.047956	1.37%
7	remaining_lease × 5-ROOM	0.046661	1.41%
8	town_QUEENSTOWN	0.040957	0.58%

#	Feature	Mean Coef	CV (%)
9	floor_area × town_PUNGGOL	0.038202	2.09%
10	town_CHOA CHU KANG	−0.037396	0.21%
11	town_BISHAN	0.036783	0.31%
12	town_WOODLANDS	−0.035789	0.73%
13	nearest_mrt_1_dist_m	−0.034882	0.90%
14	remaining_lease × 4-ROOM	0.032417	0.58%
15	gregorian_year²	0.031491	0.50%

860
Total Features

485 (56.4%)
Zeroed by Lasso

375 (43.6%)
Surviving

CV color: < 1% highly stable | 1–2% stable | > 2% moderate

Interpretation: All top-15 features have CV < 3%, confirming high cross-fold stability. gregorian_year is the dominant and most stable driver (CV = 0.47%).

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Research Question 4

Building the Best HDB Resale Price Estimator — Ensemble Learning with Bagging, Boosting, and Stacking

The Objective:

Determine if ensemble methods (Random Forest, Gradient Boosting, Stacking) significantly outperform individual models (Linear, Ridge, Lasso, Decision Tree).

Research Questions:

- Can ensemble methods (Random Forest, Gradient Boosting, Stacking) significantly outperform individual baseline models on genuinely future transactions?

- Which ensemble strategy best handles the temporal gap between historical training data and future test data?

Methodology

Dataset

224,665 HDB resale transactions enriched with 70 spatial and structural features.

Chronological Splitting

Training Set: ≤ 2023 (169,150 rows)

Validation Set: 2024 (27,832 rows)

Test Set: ≥ 2025 (27,683 rows)

Hyperparameter Tuning

5-Fold Time-Series Cross-Validation on the training set to prevent temporal data leakage.

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Model Performance

Performance on Future Data (2025+)

The Linear Baseline Ceiling

- R^2: ~0.62
- RMSE: ~$1,020 / sqm
- MAE: ~$850 / sqm

The Ensemble Breakthrough:

- R^2: 0.731
- RMSE: $865.13 / sqm
- MAE: $725.87 / sqm

Key Findings:

- Ensemble capture complex, non-linear interactions that simple models miss.
- Stacking is the Ultimate Solution for the Temporal Gap

Model	Test MAE	Test RMSE	Test R2
Linear Regression	849.92	1020.71	0.6261
Ridge	849.83	1020.56	0.6263
Lasso	839.79	1013.6	0.6313
Decision Tree	836.01	1025.2	0.6229
Random Forest	827.34	985.85	0.6513
Gradient Boosting	803.08	935.77	0.6858
Stacking Ensemble	725.87	865.13	0.7314

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   Impact of covid, cooling measures and rate hike

3. **School Proximity Premium**
   Good-deal classification via stratified median & Mutual Information analysis

4. **MRT, Storey & Lease — Price Drivers**
   Correlation analysis, EDA fly-bys & linear regression feature selection

5. **Best Price Estimator — Ensembles**
   Random Forest, Gradient Boosting, & Stacking

6. **Conclusion & Key Findings**
   Regime impact summary, model recommendations & practical implications

Conclusion on Research Findings

Myth Busted: The "Lunar Calendar" effect is a statistical illusion. Once structural controls (lease, location) are applied, lunar features provide zero predictive signal.

The Tangible Factors: Geography (Town) and Remaining Lease are the most stable, reliable predictors across all market cycles.

The Premium Proximities:

- Schools: A distinct "School Premium" exists, reliably separated by location (mature estates dominate).
- Transport: MRT proximity is a strong linear driver, though its exact premium shifted slightly during the post-COVID WFH era.

The Temporal Trap: "Year" strongly explains the past but is unreliable for the future. Time acts as a proxy for sudden market regime shifts, not a continuous trend.

Conclusion on Research Findings

- The Linear Ceiling: Traditional linear models struggle to capture complex, non-linear interactions
- The Power of Complexity: Advanced techniques are necessary to retain and utilize complex polynomial features without over-penalizing them.
- The Ultimate Solution (Ensemble Stacking): Individual models are vulnerable to the temporal gaps in real estate.
- An Optimized Stacking Architecture (blending Random Forest and Gradient Boosting) is required for more accurate forecasting.
- Result: It provides the highest resilience against future market shocks, proving that advanced machine learning is required to decode the Singaporean housing market.

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Thank You